

Predicate Logic

COMP2711 Discrete Mathematical Tools

Rosen, Sections 1.4–1.5

Mingxun Zhou (mingxunz@ust.hk)

SECTION 1

Motivation

Propositions and predicates

Propositional logic

P_A : "Alice submitted,"

P_B : "Bob submitted,"

P_C : "Carol submitted."

Propositions and predicates

Propositional logic

P_A : “Alice submitted,”

P_B : “Bob submitted,”

P_C : “Carol submitted.”

To say that every student submitted:

$$P_A \wedge P_B \wedge P_C \wedge \dots$$

We need one proposition per student.

Propositions and predicates

Propositional logic

P_A : “Alice submitted,”

P_B : “Bob submitted,”

P_C : “Carol submitted.”

To say that every student submitted:

$$P_A \wedge P_B \wedge P_C \wedge \dots$$

We need one proposition per student.

Predicate logic

Domain: all students in the class.

Let $S(x)$ mean

“ x submitted the assignment.”

Then one formula says the whole claim:

$$\boxed{\forall x S(x)}.$$

Propositions and predicates

Propositional logic

P_A : “Alice submitted,”

P_B : “Bob submitted,”

P_C : “Carol submitted.”

To say that every student submitted:

$$P_A \wedge P_B \wedge P_C \wedge \dots$$

We need one proposition per student.

The gain is expressiveness: one formula covers the whole class.

Predicate logic

Domain: all students in the class.

Let $S(x)$ mean

“ x submitted the assignment.”

Then one formula says the whole claim:

$$\boxed{\forall x S(x)}.$$

Elements of predicate logic

Domain: $\mathbb{Z}$

$\exists x \text{ Even}(x)$

“There is an even integer.”

Elements of predicate logic

domain

$\mathbb{Z}$: all integers

Domain: $\mathbb{Z}$

$\exists x \text{ Even}(x)$

“There is an even integer.”

Elements of predicate logic

domain

$\mathbb{Z}$: all integers

Domain: $\mathbb{Z}$

$\exists x \text{ Even}(x)$

"There is an even integer."

variables

x : an object name

Elements of predicate logic

domain

$\mathbb{Z}$: all integers

Domain: $\mathbb{Z}$

$\exists x \text{ Even}(x)$

"There is an even integer."

variables

x : an object name

predicate

$\text{Even}(x)$: a property of x

Elements of predicate logic



Domains

$\mathbb{Z}$ = all integers = $\{\dots, -2, -1, 0, 1, 2, \dots\}$

$\mathbb{R}$ = all real numbers.

Consider the open formula

$$x^2 \geq x.$$

Unless stated otherwise, we assume domains are nonempty.

Domains

$\mathbb{Z}$ = all integers = $\{\dots, -2, -1, 0, 1, 2, \dots\}$

$\mathbb{R}$ = all real numbers.

Consider the open formula

$$x^2 \geq x.$$

Over $\mathbb{Z}$, it is true for every x .

Unless stated otherwise, we assume domains are nonempty.

Domains

$\mathbb{Z}$ = all integers = $\{\dots, -2, -1, 0, 1, 2, \dots\}$

$\mathbb{R}$ = all real numbers.

Consider the open formula

$$x^2 \geq x.$$

Over $\mathbb{Z}$, it is true for every x .

Over $\mathbb{R}$, it is false for $x = \frac{1}{2}$.

Unless stated otherwise, we assume domains are nonempty.

Domains

$\mathbb{Z}$ = all integers = $\{\dots, -2, -1, 0, 1, 2, \dots\}$

$\mathbb{R}$ = all real numbers.

Consider the open formula

$$x^2 \geq x.$$

Over $\mathbb{Z}$, it is true for every x .

Over $\mathbb{R}$, it is false for $x = \frac{1}{2}$.

The symbols did not change. The domain did.

Unless stated otherwise, we assume domains are nonempty.

SECTION 2

Quantifiers

Universal and existential quantifiers

$$\forall x P(x)$$

$$\exists x P(x)$$

Universal and existential quantifiers

$$\forall x P(x)$$

Meaning

For all x in the domain, $P(x)$ is true.

$$\exists x P(x)$$

Meaning

There is some x in the domain such that $P(x)$ is true.

Universal and existential quantifiers

$$\forall x P(x)$$

Meaning

For all x in the domain, $P(x)$ is true.

To prove it

Prove or check $P(x)$ for all x .

$$\exists x P(x)$$

Meaning

There is some x in the domain such that $P(x)$ is true.

To prove it

Just find one x for which $P(x)$ is true.

Universal and existential quantifiers

$$\forall x P(x)$$

Meaning

For all x in the domain, $P(x)$ is true.

To prove it

Prove or check $P(x)$ for all x .

To disprove it

Find one counterexample x for which $P(x)$ is false.

$$\exists x P(x)$$

Meaning

There is some x in the domain such that $P(x)$ is true.

To prove it

Just find one x for which $P(x)$ is true.

To disprove it

Show $P(x)$ is false for all x .

Free and bound variables

Set the domain to $\mathbb{Z}$, the set of all integers.

Once the domain is fixed as $\mathbb{Z}$, we often shorten this to $\exists x \text{ Prime}(x)$.

Free and bound variables

Set the domain to $\mathbb{Z}$, the set of all integers.

Open formula

$\text{Prime}(x)$

“ x is prime.”

Its truth depends on x , so x is **free**.

Once the domain is fixed as $\mathbb{Z}$, we often shorten this to $\exists x \text{Prime}(x)$.

Free and bound variables

Set the domain to $\mathbb{Z}$, the set of all integers.

Open formula

$\text{Prime}(x)$

“ x is prime.”

Its truth depends on x , so x is **free**.

Proposition

$\underbrace{\exists x \in \mathbb{Z}}_{\text{quantifier + domain}} \underbrace{\text{Prime}(x)}_{\text{scope}}$

“There exists an integer x such that x is prime.”

Once the domain is fixed as $\mathbb{Z}$, we often shorten this to $\exists x \text{Prime}(x)$.

Free and bound variables

Set the domain to $\mathbb{Z}$, the set of all integers.

Open formula

$\text{Prime}(x)$

“ x is prime.”

Its truth depends on x , so x is **free**.

When we **bind a variable with a quantifier within a domain**, it is no longer free.
When a formula has no free variables, it is a **proposition** that is either true or false.

Once the domain is fixed as $\mathbb{Z}$, we often shorten this to $\exists x \text{Prime}(x)$.

Proposition

$\underbrace{\exists x \in \mathbb{Z}}_{\text{quantifier + domain}} \quad \underbrace{\text{Prime}(x)}_{\text{scope}}$

“There exists an integer x such that x is prime.”

Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

$\forall n$

choose every integer

Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

$$\forall n(n > 1 \rightarrow \dots)$$

restrict the requirement to integers greater than 1

Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

$$\forall n(n > 1 \rightarrow \exists p(\dots))$$

ask for a prime divisor

Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

$$\forall n \left(n > 1 \rightarrow \exists p (\text{Prime}(p) \wedge p \mid n) \right)$$

Read aloud: "For every integer n , if $n > 1$, then there exists a prime p that divides n ."

Quantifier negation

$$\neg \forall x P(x) \equiv \exists x \neg P(x),$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x).$$

Quantifier negation

$$\neg \forall x P(x) \equiv \exists x \neg P(x),$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x).$$

Not every object works $\iff$ some object fails.

Quantifier negation

$$\neg \forall x P(x) \equiv \exists x \neg P(x),$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x).$$

Not every object works $\iff$ some object fails.

No object works $\iff$ every object fails.

Negation example

Let $H(n) := \exists p(\text{Prime}(p) \wedge p \mid n)$.

Read $H(n)$: “ n has a prime divisor.”

Task: Negate $\forall n(n > 1 \rightarrow H(n))$.

In words: “Every integer greater than 1 has a prime divisor.”

Negation example

Let $H(n) := \exists p(\text{Prime}(p) \wedge p \mid n)$.

Read $H(n)$: “ n has a prime divisor.”

Task: Negate $\forall n(n > 1 \rightarrow H(n))$.

In words: “Every integer greater than 1 has a prime divisor.”

$$\neg \forall n(n > 1 \rightarrow H(n)) \equiv \exists n \neg(n > 1 \rightarrow H(n))$$

negate $\forall$: switch to $\exists$

Negation example

Let $H(n) := \exists p(\text{Prime}(p) \wedge p \mid n)$.

Read $H(n)$: “ n has a prime divisor.”

Task: Negate $\forall n(n > 1 \rightarrow H(n))$.

In words: “Every integer greater than 1 has a prime divisor.”

$$\neg \forall n(n > 1 \rightarrow H(n)) \equiv \exists n \neg(n > 1 \rightarrow H(n))$$

$$\equiv \exists n(n > 1 \wedge \neg H(n))$$

negate $\forall$: switch to $\exists$

$$\neg(A \rightarrow B) \equiv A \wedge \neg B$$

Negation example

Let $H(n) := \exists p(\text{Prime}(p) \wedge p \mid n)$.

Read $H(n)$: “ n has a prime divisor.”

Task: Negate $\forall n(n > 1 \rightarrow H(n))$.

In words: “Every integer greater than 1 has a prime divisor.”

$$\neg \forall n(n > 1 \rightarrow H(n)) \equiv \exists n \neg(n > 1 \rightarrow H(n))$$

negate $\forall$: switch to $\exists$

$$\equiv \exists n(n > 1 \wedge \neg H(n))$$

$$\neg(A \rightarrow B) \equiv A \wedge \neg B$$

$$\equiv \exists n \left(n > 1 \wedge \forall p \neg(\text{Prime}(p) \wedge p \mid n) \right)$$

expand H ; negate $\exists$

Negation example

Let $H(n) := \exists p(\text{Prime}(p) \wedge p \mid n)$.

Read $H(n)$: “ n has a prime divisor.”

Task: Negate $\forall n(n > 1 \rightarrow H(n))$.

In words: “Every integer greater than 1 has a prime divisor.”

$\neg \forall n(n > 1 \rightarrow H(n)) \equiv \exists n \neg(n > 1 \rightarrow H(n))$	negate $\forall$: switch to $\exists$
$\equiv \exists n(n > 1 \wedge \neg H(n))$	$\neg(A \rightarrow B) \equiv A \wedge \neg B$
$\equiv \exists n \left(n > 1 \wedge \forall p \neg(\text{Prime}(p) \wedge p \mid n) \right)$	expand H ; negate $\exists$
$\equiv \exists n \left(n > 1 \wedge \forall p (\neg \text{Prime}(p) \vee p \nmid n) \right)$	$\neg(A \wedge B) \equiv \neg A \vee \neg B$

Negation example

Let $H(n) := \exists p(\text{Prime}(p) \wedge p \mid n)$.

Read $H(n)$: “ n has a prime divisor.”

Task: Negate $\forall n(n > 1 \rightarrow H(n))$.

In words: “Every integer greater than 1 has a prime divisor.”

$\neg \forall n(n > 1 \rightarrow H(n)) \equiv \exists n \neg(n > 1 \rightarrow H(n))$	negate $\forall$: switch to $\exists$
$\equiv \exists n(n > 1 \wedge \neg H(n))$	$\neg(A \rightarrow B) \equiv A \wedge \neg B$
$\equiv \exists n \left(n > 1 \wedge \forall p \neg(\text{Prime}(p) \wedge p \mid n) \right)$	expand H ; negate $\exists$
$\equiv \exists n \left(n > 1 \wedge \forall p (\neg \text{Prime}(p) \vee p \nmid n) \right)$	$\neg(A \wedge B) \equiv \neg A \vee \neg B$
$\equiv \exists n \left(n > 1 \wedge \forall p (\text{Prime}(p) \rightarrow p \nmid n) \right)$	$\neg A \vee B \equiv A \rightarrow B.$

Negation example

Let $H(n) := \exists p(\text{Prime}(p) \wedge p \mid n)$.

Read $H(n)$: “ n has a prime divisor.”

Task: Negate $\forall n(n > 1 \rightarrow H(n))$.

In words: “Every integer greater than 1 has a prime divisor.”

$\neg \forall n(n > 1 \rightarrow H(n)) \equiv \exists n \neg(n > 1 \rightarrow H(n))$	negate $\forall$: switch to $\exists$
$\equiv \exists n(n > 1 \wedge \neg H(n))$	$\neg(A \rightarrow B) \equiv A \wedge \neg B$
$\equiv \exists n \left(n > 1 \wedge \forall p \neg(\text{Prime}(p) \wedge p \mid n) \right)$	expand H ; negate $\exists$
$\equiv \exists n \left(n > 1 \wedge \forall p (\neg \text{Prime}(p) \vee p \nmid n) \right)$	$\neg(A \wedge B) \equiv \neg A \vee \neg B$
$\equiv \exists n \left(n > 1 \wedge \forall p (\text{Prime}(p) \rightarrow p \nmid n) \right)$	$\neg A \vee B \equiv A \rightarrow B$.

In English: **some integer greater than 1 has no prime divisor**. This is false.

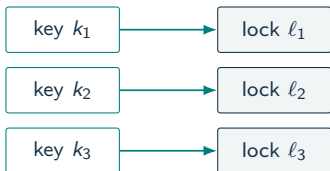
SECTION 3

Quantifier order

Quantifier-order fallacy

Let $\text{Opens}(k, \ell)$ mean “key k opens lock ℓ .”

Every lock has some key

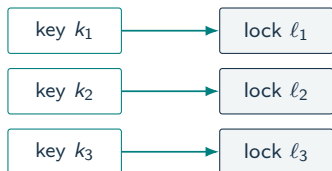


$$\forall \ell \exists k \text{ Opens}(k, \ell)$$

Quantifier-order fallacy

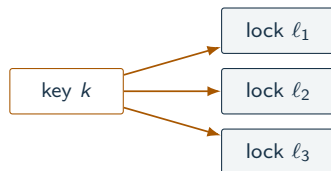
Let $\text{Opens}(k, \ell)$ mean “key k opens lock ℓ .”

Every lock has some key



$$\forall \ell \exists k \text{ Opens}(k, \ell)$$

One key opens every lock



$$\exists k \forall \ell \text{ Opens}(k, \ell)$$

Reordering quantifiers

Same type: swapping preserves meaning

$$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$$

“ P holds for every pair.”

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

“ P holds for some pair.”

Reordering quantifiers

Same type: swapping preserves meaning

$$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$$

“P holds for every pair.”

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

“P holds for some pair.”

Mixed types: one direction is always valid

$$\exists y \forall x R(x, y) \implies \forall x \exists y R(x, y)$$

“One y works for every x” implies “for every x, some y works.”

Reordering quantifiers

Same type: swapping preserves meaning

$$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$$

“P holds for every pair.”

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

“P holds for some pair.”

Mixed types: one direction is always valid

$$\exists y \forall x R(x, y) \implies \forall x \exists y R(x, y)$$

“One y works for every x” implies “for every x, some y works.”

The reverse direction is generally false

$$\forall x \exists y R(x, y) \not\Rightarrow \exists y \forall x R(x, y)$$

“Each x may have its own y” does not imply “one y works for every x.”

SECTION 4

Translation

Translation method

1. **State the domain.**
2. **Define predicates** and their argument order.
3. **Determine the choices:** what is selected first?
4. **Write the formula:** apply restrictions correctly.
5. **Read back** and test a small example.

Translation is not word replacement. It is reconstruction of the sentence's logic.

Existence and uniqueness

Translate: **Every even integer has exactly one integer half.**

Existence and uniqueness

Translate: **Every even integer has exactly one integer half.**

Existence for a fixed n

$$\exists k (n = 2k)$$

Existence and uniqueness

Translate: **Every even integer has exactly one integer half.**

Uniqueness after choosing k

$$\forall j (n = 2j \rightarrow j = k)$$

Existence and uniqueness

Translate: **Every even integer has exactly one integer half.**

$$\forall n (\text{Even}(n) \rightarrow \exists k (n = 2k \wedge \forall j (n = 2j \rightarrow j = k)))$$

Existence and uniqueness

Translate: **Every even integer has exactly one integer half.**

$$\forall n (\text{Even}(n) \rightarrow \exists k (n = 2k \wedge \forall j (n = 2j \rightarrow j = k)))$$

One such value exists, and any value that works must be that same value. The notation $\exists!k$ abbreviates this pattern.

Mathematical writing

In papers and textbooks, writers rarely stay at either extreme. Consider Goldbach's conjecture, with all variables ranging over $\mathbb{Z}$.

Fully verbal

Every even integer greater than two is the sum of two prime numbers.

Mathematical writing

In papers and textbooks, writers rarely stay at either extreme. Consider Goldbach's conjecture, with all variables ranging over $\mathbb{Z}$.

Fully verbal

Every even integer greater than two is the sum of two prime numbers.

Fully symbolic

$$\forall n \left((\text{Even}(n) \wedge n > 2) \rightarrow \exists p \exists q (\text{Prime}(p) \wedge \text{Prime}(q) \wedge n = p + q) \right).$$

Mathematical writing

In papers and textbooks, writers rarely stay at either extreme. Consider Goldbach's conjecture, with all variables ranging over $\mathbb{Z}$.

Fully verbal

Every even integer greater than two is the sum of two prime numbers.

Fully symbolic

$$\forall n \left((\text{Even}(n) \wedge n > 2) \rightarrow \exists p \exists q (\text{Prime}(p) \wedge \text{Prime}(q) \wedge n = p + q) \right).$$

Typical mathematical prose

Every even integer $n > 2$ can be written as $n = p + q$ for primes p and q .

Mathematical writing

In papers and textbooks, writers rarely stay at either extreme. Consider Goldbach's conjecture, with all variables ranging over $\mathbb{Z}$.

Fully verbal

Every even integer greater than two is the sum of two prime numbers.

Fully symbolic

$$\forall n \left((\text{Even}(n) \wedge n > 2) \rightarrow \exists p \exists q (\text{Prime}(p) \wedge \text{Prime}(q) \wedge n = p + q) \right).$$

Typical mathematical prose

Every even integer $n > 2$ can be written as $n = p + q$ for primes p and q .

The usual register uses **words for meaning** and **notation for structure**.

Translation practice

Domain: the integers.

Some prime divides every even integer.

Translation practice

Domain: the integers.

Some prime divides every even integer.

$$\exists p \left(\text{Prime}(p) \wedge \forall n (\text{Even}(n) \rightarrow p \mid n) \right).$$

Translation practice

Domain: the integers.

Some prime divides every even integer.

$$\exists p \left(\text{Prime}(p) \wedge \forall n (\text{Even}(n) \rightarrow p \mid n) \right).$$

Choose $p = 2$, then check an arbitrary even integer n .

SECTION 5

Equivalence and normal form

Valid distribution laws

Universal quantifier with “and”

$$\forall x (P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$$

“Every object has both properties P and Q ” means “every object has P , and every object has Q .”

Valid distribution laws

Universal quantifier with “and”

$$\forall x (P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$$

“Every object has both properties P and Q” means “every object has P, and every object has Q.”

Existential quantifier with “or”

$$\exists x (P(x) \vee Q(x)) \equiv (\exists x P(x)) \vee (\exists x Q(x))$$

“Some object has property P or Q” means “some object has P, or some object has Q.”

Universal quantification and disjunction

Always valid

$$(\forall x P(x)) \vee (\forall x Q(x)) \implies \forall x (P(x) \vee Q(x))$$

If one property holds for every object, then every object has at least one of the two properties.

Universal quantification and disjunction

The converse can fail

Domain: all integers.

Even integers: $\dots, -2, 0, 2, 4, \dots$

Odd integers: $\dots, -3, -1, 1, 3, \dots$

True in this model

$$\forall x (\text{Even}(x) \vee \text{Odd}(x))$$

“Every integer is even or odd.”



False in this model

$$(\forall x \text{ Even}(x)) \vee (\forall x \text{ Odd}(x))$$

“Every integer is even, or every integer is odd.”

Existential quantification and conjunction

Always valid

$$\exists x (P(x) \wedge Q(x)) \implies (\exists x P(x)) \wedge (\exists x Q(x))$$

If one object has both properties, then each separate existence claim is true.

Existential quantification and conjunction

The converse can fail

Domain: all integers.

Even integers: $\dots, -2, 0, 2, 4, \dots$

Odd integers: $\dots, -3, -1, 1, 3, \dots$

True in this model

$$(\exists x \text{ Even}(x)) \wedge (\exists x \text{ Odd}(x))$$

"Some integer is even; some integer is odd."



False in this model

$$\exists x (\text{Even}(x) \wedge \text{Odd}(x))$$

"Some integer is both even and odd."

Prenex normal form

Prenex normal form has all quantifiers in front:

$$Q_1x_1 \ Q_2x_2 \ \cdots \ Q_nx_n \ M, \quad M \text{ has no quantifiers.}$$

Prenex normal form

Prenex normal form has all quantifiers in front:

$$Q_1x_1 Q_2x_2 \cdots Q_nx_n M, \quad M \text{ has no quantifiers.}$$

1. Rename bound variables to distinct names.
2. Eliminate $\rightarrow$ and $\leftrightarrow$.
3. Move negations inward, switching crossed quantifiers.
4. Pull quantifiers outward.

Prenex normal form

Prenex normal form has all quantifiers in front:

$$Q_1x_1 \ Q_2x_2 \ \cdots \ Q_nx_n \ M, \quad M \text{ has no quantifiers.}$$

1. Rename bound variables to distinct names.
2. Eliminate $\rightarrow$ and $\leftrightarrow$.
3. Move negations inward, switching crossed quantifiers.
4. Pull quantifiers outward.

Pull a quantifier past another part only when its variable is not free there. Do one justified transformation per line.

PNF conversion

Convert

$$\exists z \left(\exists y R(y, z) \rightarrow \forall y Q(y, z) \right)$$

to prenex normal form.

PNF conversion

Convert

$$\exists z \left(\exists y R(y, z) \rightarrow \forall y Q(y, z) \right)$$

to prenex normal form.

The two y quantifiers have different scopes. Rename one first:

PNF conversion

Convert

$$\exists z \left(\exists y R(y, z) \rightarrow \forall y Q(y, z) \right)$$

to prenex normal form.

The two y quantifiers have different scopes. Rename one first:

$$\exists z \left(\exists y R(y, z) \rightarrow \forall x Q(x, z) \right).$$

PNF conversion steps

$$\exists z \left(\exists y R(y, z) \rightarrow \forall x Q(x, z) \right)$$

PNF conversion steps

$$\exists z \left(\exists y R(y, z) \rightarrow \forall x Q(x, z) \right)$$

$$\equiv \exists z \left(\neg \exists y R(y, z) \vee \forall x Q(x, z) \right)$$

remove the arrow

PNF conversion steps

$$\exists z \left(\exists y R(y, z) \rightarrow \forall x Q(x, z) \right)$$

$$\equiv \exists z \left(\neg \exists y R(y, z) \vee \forall x Q(x, z) \right)$$

remove the arrow

$$\equiv \exists z \left(\forall y \neg R(y, z) \vee \forall x Q(x, z) \right)$$

move negation inward

PNF conversion steps

$$\exists z \left(\exists y R(y, z) \rightarrow \forall x Q(x, z) \right)$$

$$\equiv \exists z \left(\neg \exists y R(y, z) \vee \forall x Q(x, z) \right)$$

$$\equiv \exists z \left(\forall y \neg R(y, z) \vee \forall x Q(x, z) \right)$$

$$\equiv \boxed{\exists z \forall y \forall x (\neg R(y, z) \vee Q(x, z))}$$

remove the arrow

move negation inward

pull quantifiers outward.

PNF conversion steps

$$\exists z \left(\exists y R(y, z) \rightarrow \forall x Q(x, z) \right)$$

$$\equiv \exists z \left(\neg \exists y R(y, z) \vee \forall x Q(x, z) \right)$$

remove the arrow

$$\equiv \exists z \left(\forall y \neg R(y, z) \vee \forall x Q(x, z) \right)$$

move negation inward

$$\equiv \boxed{\exists z \forall y \forall x (\neg R(y, z) \vee Q(x, z))}$$

pull quantifiers outward.

The remaining formula $\neg R(y, z) \vee Q(x, z)$ has no quantifiers.

Summary

1. Predicates describe properties and relations over a stated domain.
 2. Quantifiers bind variables and determine the evidence a claim needs.
 3. Quantifier order determines whether one choice may depend on another; scope controls meaning.
 4. Negation, counterexamples, and PNF transformations must preserve that structure.
-

Next lecture

Inference and proof: deriving conclusions from quantified premises.